Calibo AI Academy

BUSINESS INTELLIGENCE CASE STUDY

Power BI Sales Intelligence

Sales Performance, Profitability & Business Intelligence Dashboard — use case, solution, approach and expected outcome

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USE CASE TITLE

Sales Performance, Profitability & Business Intelligence Dashboard

A Power BI solution that turns three years of scattered sales, product, customer, salesperson, location and budget data into one interactive dashboard — so management can see where revenue and profit are coming from, and where to focus attention, in seconds instead of days.

1 The Problem in Simple Words

Data is scattered

2021, 2022 and 2023 sales sit in three separate tables, alongside separate product, location, customer, salesperson and budget files.

Reports are manual

Managers rebuild Excel reports by hand to compare years, products or locations — slow, error-prone, and never quite up to date.

Insight is hidden

Without a connected model, it's hard to see which products, customers, salespeople or locations are actually driving profit — or missing budget.

2 Our Solution: A Connected Sales Model

Think of it as one control room that cleans the data, models it properly, scores performance with DAX, and lets any manager click through from company-wide results down to a single order. Six-stage build:

STAGE 1
Prepare
Clean 2021–2023 sales in Power Query, append into one Fact_Sales table, fix types and blanks.

STAGE 2
Model
Build a star schema: Fact_Sales linked to Dim_Product, Dim_Location, Dim_Customer, Dim_Salesperson.

STAGE 3
Date Table
Add a marked Date dimension with Year, Quarter, Month (chronological), Week and Day.

STAGE 4
Measure
Write DAX for revenue, cost, profit, margin, YoY growth, YTD/MTD and budget variance.

STAGE 5
Visualize
Build 7 report pages: Executive Overview, Trend, Product, Customer, Salesperson, Geography, Budget.

STAGE 6
Interact
Add slicers, drill-down/drill-through, tooltips and cross-filtering across every page.

3 Understanding the Data (Two Charts)

Based on the full dataset: 10,889 orders across 2021–2023 (2023 is a partial, 2.5-month slice).

Chart 1 — Revenue by Year

Which year is carrying the business?



- 2021 — $11.69M (45.6%)
- 2022 — $11.57M (45.1%)

Chart 2 — Cost vs. Gross Profit

How much of every sale is margin?



- Total Cost — $17.32M (67.5%)
- Gross Profit — $8.34M (32.5%)

2023* — $2.40M (9.4%)

*2023 covers only Jan 1 – Mar 15, so its slice is naturally small — not a sales drop.

A steady ~32.5% margin across 101 products gives management a clear baseline to track drift by product or location.

4 Procedure I Followed (Approach)

1. **Load and understand the data**
   Imported 9 tables: 3 yearly sales sheets, Products, Locations, Customers, Sales People, Budgeting, Date Ranges.
2. **Profile and clean**
   Checked types, blanks and duplicates; found 23 unmatched Location IDs in Budgeting and fixed monthly allocation columns.
3. **Append and shape the fact table**
   Combined 2021–2023 sales (10,889 rows) into one Fact_Sales table with consistent columns and types.
4. **Build the star schema**
   Related Fact_Sales to four dimension tables on their ID keys — no many-to-many, no snowflaking.
5. **Build the Date table and DAX measures**
   Marked an official Date table, then wrote sales, cost/profit, time-intelligence and budget measures.
6. **Design the seven dashboard pages**
   Executive Overview, Sales Trend, Product, Customer, Salesperson, Geographic and Budget vs Actual.
7. **Add interactivity and validate**
   Slicers, drill-down, drill-through and tooltips added; totals cross-checked against source tables before sign-off.

Power Query | Star Schema Modeling | DAX | Time Intelligence | Power BI Desktop

5 Three Design Principles

1. One source of truth
Every visual reads from the same star schema, so numbers never disagree page to page.

2. Measures, not manual math
Revenue, cost, margin and YoY growth are DAX measures — never hard-typed numbers.

3. Every page is interactive
Slicers, drill-through and tooltips let any manager self-serve their own answer.

6 Expected Outcome

Results measured directly from the provided 2021–2023 dataset:

$25.66M Total Revenue	**$8.34M** Gross Profit (32.5%)	**10,889** Total Orders	**21,078** Units Sold
101 Products Tracked	**74** Locations Tracked	**801** Customers	**45** Salespeople

✓ **Faster decisions:** Executive KPIs replace a manual Excel rebuild with a live, one-click view.

✓ **Clear top performers:** Rialto, Roseville and Victorville lead all 74 locations by revenue.

✓ **Margin visibility:** Product-level margins range from ~15% to ~50%, exposing pricing opportunities.

✓ **Concentration risk flagged:** the top 10 customers contribute only ~5.4% of revenue — a broad, low-risk base worth confirming with the business.

✓ **Budget accountability:** location-level Budget vs Actual view shows exactly who is over or under target each month.

7 Business Intelligence Concepts Used

Concept	How it is used in this project
Data ingestion	2021–2023 sales, product, location, customer, salesperson and budget tables imported via Power Query.
Data quality	Type checks, blank/null handling, duplicate checks, and reconciling mismatched Budgeting location IDs.

Transformation	Appending yearly sales tables, renaming columns, and creating calculated columns (e.g. Revenue, Profit).
Data modeling	Star schema — one Fact_Sales table related to four dimension tables plus a marked Date table.
DAX & time intelligence	CALCULATE, FILTER, DIVIDE, SUMX, SAMEPERIODLASTYEAR, TOTALYTD, RANKX for growth, YTD and rankings.
Interactivity	Slicers, drill-down (Year→Quarter→Month→Day), drill-through pages, and cross-filtering.
Reporting & analytics	7 dashboard pages covering executive, trend, product, customer, salesperson, geography and budget views.

8 Conclusion

This project turns three years of disconnected sales, product, customer, salesperson, location and budget data into a single Power BI model that management can question directly — by year, product, customer, salesperson, city or month — without waiting on a manual report. With a clean star schema and a consistent set of DAX measures, every number on every page ties back to the same source of truth, and the same design can extend to new years, new products or new locations as the business grows.

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